

SENTINEL-CPS

System Architecture Document (SAD)

Securing Cyber Physical Systems in a Remote IoT Lab: A Zero-Trust Edge Gateway Architecture Utilizing eBPF and AI Threat Detection

Architecture Version	10.0
Author	Tarique Chowdhury
Advisor	Dr. George Li
Date	May 29, 2026
Document Status	Draft for thesis development and advisor review
Primary Artifact	Zero-Trust edge gateway architecture for remote CPS education

Prepared for Master's Thesis Development

Document Control

This document consolidates the current v10.0 Sentinel-CPS architecture into a formal system architecture specification suitable for thesis development, implementation planning, and advisor review.

Version	Date	Prepared By	Summary of Changes
10.0	May 29, 2026	Tarique Chowdhury	Formalized TV-as-a-Track execution model; updated Flask C2 application to TCP Port 8080; clarified eBPF serial tracing; replaced camera-dependent navigation with light-sensor/PID control; added USB 3.0 boot-media resilience for high-volume logging.

Contents

1. Executive Summary and Scope
2. Architectural Objectives and Design Principles
3. High-Level Architectural Topology
4. Hardware Architecture and Resilience
5. Software, Network, and Command Pipeline
6. TV-as-a-Track Cyber-Physical Interface
7. Advanced Threat Mitigation with eBPF and AI
8. Security Boundaries and Operational Assumptions
9. End-to-End Operational Workflow
10. Conclusion
11. Appendix A: Port, Interface, and Protocol Summary

1. Executive Summary and Scope

Sentinel-CPS is a secure, autonomous Cyber-Physical System (CPS) testbed designed to expose advanced robotics, IoT networking, and security monitoring concepts within a remote educational laboratory. The architecture merges a Zero-Trust gateway infrastructure with a dynamic TV-as-a-Track physical interface. Remote users can command, observe, and evaluate autonomous edge vehicles through a unified digital twin while the physical devices remain constrained behind hardened network, host, and hardware boundaries.

The scope of this System Architecture Document (SAD) includes the physical topology, gateway execution model, network segmentation strategy, telemetry and command pipeline, TV-based track generation model, and planned eBPF/AI threat monitoring layer. The document is intended to guide implementation, testing, and thesis validation rather than serve as a general product specification.

2. Architectural Objectives and Design Principles

- **Zero-Trust exposure:** The gateway is treated as the only authorized mediator between remote users and physical CPS devices. No edge node receives direct enterprise-network trust.
- **Remote educational usability:** Authorized students and reviewers can interact with the system through a controlled Bastion Host and web-based Command and Control (C2) interface.
- **Bare-metal observability:** The Gateway runs natively under systemd, avoiding container abstraction so eBPF tools can trace host-level syscalls and serial I/O directly.
- **Hardware-constrained safety:** Control commands pass through a tethered ESP32 translation Hub and can be interrupted by emergency STOP logic before unsafe physical behavior escalates.
- **Dynamic physical experimentation:** A Smart TV renders the driving environment as pixels, allowing rapid track changes without physical tape, printing, or mechanical reconfiguration.
- **Defense in depth:** MAC whitelisting, Ed25519 SSH enforcement, nftables policy, serial mediation, and planned anomaly detection reinforce one another rather than relying on a single control.

3. High-Level Architectural Topology

The Sentinel-CPS ecosystem is segmented into four operational layers. Each layer has a distinct role, trust posture, and enforcement boundary.

Layer	Primary Role	Core Components	Trust Boundary
Remote Educational Sandbox / Ingress	Provides sanctioned remote access for students, reviewers, and administrators.	University VPN, Windows Bastion Host, RDP session, isolated development microcontrollers.	Only the Bastion Host is allowed to bridge remote users into the lab workflow.
Zero-Trust Gateway / Overwatch	Mediates all CPS control, logging, web access, and kernel-level monitoring.	Raspberry Pi 4, Linux, nftables HLAC, Flask C2 app, systemd service, eBPF tooling.	Default-deny inbound policy; key-based administration only; host-level observability.
Edge Network / Encrypted Telemetry	Carries operational telemetry and command packets between the Gateway Hub and autonomous vehicles.	Tethered ESP32-D0WDQ6 Hub, ESP-NOW, CCMP encryption, edge ESP32 vehicles.	Gateway never directly trusts open wireless traffic; telemetry is mediated by the Hub.
Dynamic Physical Substrate / Execution	Provides the live road surface and visual state representation for the CPS experiment.	Horizontally mounted Smart TV, Lane Builder output, digital twin overlay, light-sensor-readable track.	Display endpoint is authorized only for its required web/display function.

4. Hardware Architecture and Resilience

The hardware architecture is designed to minimize avoidable failure modes while preserving the security properties needed for research validation. The Gateway, Hub, edge vehicles, Bastion Host, and Display Node are separated by function so that compromise or instability in one layer does not automatically grant unrestricted control of the CPS.

Component	Function	Implementation Notes	Security / Resilience Rationale
Gateway Node: Raspberry Pi 4	Central enforcement node and C2 host.	Runs natively under systemd without Docker. Boots and operates from a high-throughput USB 3.0 flash drive rather than MicroSD.	Avoids MicroSD wear from high-volume eBPF/application logging and preserves host-level syscall visibility.
Hardware Translation Hub: ESP32-D0WDQ6	Serial-to-wireless bridge between Gateway and edge vehicles.	Physically tethered to the Gateway high-power USB 3.0 bus at /dev/ttyUSB0.	Mitigates prior USB power-sag risk and prevents the Gateway from directly processing raw wireless control traffic.
Autonomous Edge Nodes: ESP32 Vehicles	Physical CPS agents executing track-following behavior.	Chassis include downward-facing analog light sensors such as LDRs or phototransistors, motor drivers, and onboard PID logic.	Replaces camera-dependent visual payloads to reduce latency, bandwidth demand, and video spoofing surface.
Bastion Host: Windows Workstation	Sanctioned ingress point for remote educational users.	Reached through University VPN and RDP. Used for controlled web interaction and development workflow access.	Prevents direct off-campus access to the Gateway and concentrates remote access auditing at a known endpoint.
Display Node: Smart TV	Physical track substrate and digital twin visualization endpoint.	Mounted horizontally to emit the track surface and display the telemetry overlay through the web application.	Permitted only for required display/web behavior; not treated as an administrative endpoint.

5. Software, Network, and Command Pipeline

5.1 Local Resolution and Hardware-Level Access Control

The architecture avoids dependency on static DHCP reservations by using avahi-daemon for mDNS resolution through the hostname `iot-pi.local`. Inbound network traffic is governed by nftables using a default-deny model. The firewall applies Hardware-Level Access Control (HLAC) by restricting service access to explicitly approved MAC addresses.

- **SSH management:** TCP Port 22 is restricted to authorized administrative endpoints and additionally protected by Ed25519 key-only authentication.
- **C2 web application:** TCP Port 8080 hosts the unified Flask interface and is limited to authorized web clients such as the Bastion Host and Display Node.
- **Display segregation:** The Smart TV is treated as a non-administrative endpoint. It should not receive SSH privileges even when it is permitted to access the web dashboard.

5.2 Unified C2 Web Application

The unified Command and Control (C2) web application is implemented as a Python Flask service managed by `sentinel.service`. It exposes the Lane Builder, Digital Twin telemetry overlay, command dispatch functions, and logging surfaces through TCP Port 8080. Running the service natively on the host preserves direct access to serial I/O behavior for future eBPF-based observability.

5.3 Cryptographic Authentication

Password authentication is globally disabled at the SSH daemon level. Administrative access requires authorized Ed25519 cryptographic key pairs. This design prevents MAC whitelisting from becoming the sole enforcement mechanism: even if a network identity is spoofed or a trusted adapter is misused, the attacker still lacks the private key required for Gateway administration.

5.4 Command and Telemetry Flow

1. **Remote user ingress:** The user authenticates through the University VPN and reaches the Windows Bastion Host via RDP.
2. **C2 interaction:** The user opens the Gateway web application on TCP Port 8080 and configures the track, view, or control action.

3. **Gateway mediation:** The Flask service validates the requested action, records it in application logs, and sends approved commands through the serial interface.
4. **Hub translation:** The tethered ESP32-D0WDQ6 Hub receives the serial command and translates it into the encrypted ESP-NOW edge network.
5. **Edge execution:** The autonomous ESP32 vehicles interpret commands, follow the emitted track using local PID logic, and return telemetry through the Hub.
6. **Digital twin update:** The Gateway parses returned telemetry and updates the web overlay presented to the authorized display endpoint.

6. TV-as-a-Track Cyber-Physical Interface

The principal architectural advancement in Version 10.0 is the synchronization of the digital C2 interface and the physical driving environment. Instead of using static tape or printed lanes on a lab floor, a Smart TV is mounted horizontally and used as the light-emitting physical substrate for autonomous vehicle navigation.

Stage	Description	Result
Generation	The remote user uses the Lane Builder interface to generate a digital track with high-luminance lane pixels on a low-luminance background.	Track layout can be changed instantly through software.
Execution	The ESP32 vehicles are placed directly on the TV glass. Downward-facing analog light sensors read pixel luminance differences from the screen.	Onboard PID control keeps each vehicle centered over the generated lane.
Synchronization	The same web application that renders the track also overlays telemetry and operational status.	The C2 interface becomes a physical reality generator and a digital twin surface at the same time.
Validation	Track changes, telemetry updates, and control commands can be observed in one closed architecture.	The system supports repeatable CPS experiments without mechanical intervention.

This approach reduces setup time, removes the need for manual track reconstruction, and creates a repeatable physical interface for testing control, safety, and anomaly detection behavior.

7. Advanced Threat Mitigation with eBPF and AI

Sentinel-CPS incorporates host-level observability to detect attacks or unsafe behavior that may not be visible through application logs alone. The planned monitoring layer combines eBPF syscall tracing, baseline modeling, and automated mitigation.

- **eBPF serial tracing:** Using bcc-tools, the Linux kernel traces `sys_enter_write` and `sys_enter_read` activity associated with the `/dev/ttyUSB0` serial interface.
- **Behavioral baseline:** Kernel telemetry, command logs, PID steering behavior, and expected ESP-NOW payload frequency are used to model nominal system behavior.
- **AI inference:** A localized anomaly detection module evaluates live telemetry against the learned baseline to identify data injection, replay patterns, DoS conditions, or abnormal command frequency.
- **Automated perimeter severance:** When a critical anomaly is detected, the Gateway transmits an emergency STOP command and temporarily locks or severs the serial read/write path to prevent unsafe edge execution.

This mitigation path is designed to operate below the web-application layer. If the Flask application, student workflow, or wireless bridge begins producing abnormal behavior, the Gateway can still enforce a safety stop through host-level control of the serial chokepoint.

8. Security Boundaries and Operational Assumptions

Boundary / Assumption	Design Treatment	Validation Implication
Remote users are untrusted by default.	Remote activity enters through University VPN and the Windows Bastion Host rather than direct Gateway exposure.	Testing should verify that the Gateway cannot be reached directly from unapproved hosts.
The Smart TV is not an administrator.	The Display Node may access the C2 display surface but should not be whitelisted for SSH.	Firewall tests should distinguish Port 8080 display access from Port 22 management access.
Edge vehicles cannot be fully trusted.	Vehicles receive commands through the Hub and return telemetry through controlled channels.	Malformed or high-frequency telemetry should trigger logging and later anomaly scoring.
Wireless traffic is exposed to spoofing attempts.	ESP-NOW telemetry and commands use CCMP encryption and approved peer identity configuration.	Firmware testing should include unauthorized-peer rejection.
High-volume logging can stress storage.	Gateway operation uses USB 3.0 storage instead of MicroSD for improved write endurance.	Long-duration tests should monitor storage health and logging throughput.
Safety requires a chokepoint.	The serial Hub path remains the enforcement point for emergency STOP and severance logic.	Attack simulations should confirm the STOP path works under load.

9. End-to-End Operational Workflow

- 1. Authenticate remotely:** The operator connects through the University VPN and enters the Windows Bastion Host via RDP.
- 2. Open the C2 interface:** The operator accesses the Gateway web service on TCP Port 8080 from an authorized endpoint.
- 3. Generate a digital track:** The Lane Builder renders a high-contrast lane pattern on the Smart TV surface.
- 4. Deploy the edge vehicles:** ESP32 vehicles are placed on the TV substrate, where downward-facing sensors read lane luminance.
- 5. Execute autonomous navigation:** Vehicles use local PID control to follow the generated track while the Hub mediates telemetry and command flow.
- 6. Monitor the digital twin:** The web interface displays telemetry and status overlays for operational visibility.
- 7. Trace and score behavior:** The Gateway collects application logs and planned eBPF syscall telemetry for anomaly detection.
- 8. Mitigate unsafe activity:** When critical anomalies occur, the Gateway issues STOP and locks the serial pathway to protect the CPS.

10. Conclusion

Version 10.0 positions Sentinel-CPS as a controlled, research-ready CPS architecture that combines remote educational access with rigorous security boundaries. The design uses a Raspberry Pi Gateway as the central policy enforcement point, an ESP32 Hub as the serial-to-wireless mediator, ESP32 vehicles as constrained autonomous edge nodes, and a Smart TV as both the dynamic physical track and digital twin surface.

The resulting architecture supports repeatable experimentation in remote robotics and IoT security while maintaining Zero-Trust principles, kernel-level observability, and a clear safety chokepoint for automated mitigation.

Appendix A: Port, Interface, and Protocol Summary

Element	Value	Purpose	Security Treatment
---------	-------	---------	--------------------

Element	Value	Purpose	Security Treatment
Gateway hostname	iot-pi.local	Stable local access without static DHCP reservation.	Resolved via avahi-daemon/mDNS within the controlled lab context.
SSH	TCP Port 22	Administrative management.	Restricted to approved admin endpoints; Ed25519 key-only authentication.
C2 Web Application	TCP Port 8080	Lane Builder, digital twin, command workflow, and display interface.	Restricted to approved web/display endpoints through nftables HLAC.
Serial interface	/dev/ttyUSB0	Gateway-to-Hub command and telemetry chokepoint.	Observed with eBPF and subject to STOP/severance mitigation.
Hub link	USB 3.0 bus	Physical tether from Raspberry Pi to ESP32-D0WDQ6 Hub.	Limits direct Gateway exposure to wireless traffic.
Edge network	ESP-NOW with CCMP	Low-latency telemetry and motor command exchange.	Encrypted peer-to-peer communication with authorized nodes.
Physical substrate	Horizontal Smart TV	Dynamic lane rendering and digital twin visualization.	Authorized for display/web function only; not an admin endpoint.
Gateway storage	USB 3.0 flash drive	Boot media and high-volume logging target.	Reduces MicroSD wear risk under eBPF/application logging load.